

Sathvika Pathuri

DATA ANALYST

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SUMMARY

- Data Analyst with 3 years of proven expertise in architecting ETL workflows using Python (Pandas) and SQL to ingest, transform, and load complex data from ERP and claims systems into cloud-based warehouses.
- Skilled in leveraging AWS services (Redshift, Glue, Lambda) and Snowflake to build scalable data pipelines that enable near-real-time analytics and reliable data availability.
- Proficient in designing interactive dashboards and reports in Tableau and Power BI, translating multifaceted datasets into clear, actionable insights for finance and healthcare stakeholders.
- Experienced in applying statistical methods and Scikit-learn models for predictive analytics and anomaly detection, ensuring data-driven decision making and regulatory compliance.

TECHNICAL SKILLS

Methodologies:	SDLC, Agile, Waterfall
Language:	Python, SQL, R
Packages:	Pandas, NumPy, Matplotlib, SciPy, Seaborn, Scikit-Learn
Data Visualization Tools:	Tableau, Power BI, Advanced Excel, Statistics, Qlik
Database:	MySQL, MongoDB, PostgreSQL, Oracle, SQL Server
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Cloud Technologies:	Amazon Web Services (AWS), Snowflake
Other Technical Skills:	Machine Learning Algorithms, SSIS, SSRS, SAS, Mathematics, Probability distributions, ANOVA, Hypothesis Testing, Regression Analysis, Linear Algebra, Advance Analytics, Data Mining, Data warehousing, Data transformation, Data Storytelling, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, Requirement Gathering, Jira, Git, GitHub
Operating System:	Windows, Linux, Mac OS

PROFESSIONAL EXPERIENCE

Data Analyst | Aetna, CT

Aug 2024 – Present

- Engaged cross-functional stakeholders (clinicians, pharmacists, fraud investigators, and business analysts) to gather and document detailed requirements, translating business needs into technical specifications for the fraud detection platform.
- Integrated Electronic Health Records (EHR) and pharmacy claims data into a centralized AWS Redshift platform, enabling efficient cross-source analytics and reducing query execution time by 40%.
- Cleaned and transformed large volumes of prescription data using Python (Pandas) and SQL, handling missing values and inconsistencies, which led to a 20% improvement in machine learning model accuracy.
- Developed real-time ETL pipelines with AWS Lambda and AWS Glue to ingest and process over 1 million daily records, reducing manual intervention and ensuring low-latency data availability for analytics.
- Built and deployed classification models (Random Forest, SVM) for prescription fraud detection, incorporating domain-specific features, that uncovered approximately \$250 K in potential fraudulent claims and improved detection accuracy by 20%.
- Created a Tableau dashboard to visualize fraud trends, model performance, and regional outliers in real time, supporting decision-making and contributing to an 11% drop in fraudulent prescriptions.
- Optimized AWS infrastructure by tuning Redshift queries, managing distribution styles, and implementing automated storage policies, ensuring scalability and cost-effective processing of over 2 TB of monthly data.

Data Analyst | SoftAge Group, India

Jul 2021 – Jul 2023

- Supported the “Financial Performance Automation” initiative by developing Python ETL scripts with Pandas to ingest Oracle ERP general ledger entries and CSV feeds into a Snowflake warehouse, reducing manual data consolidation time by 55%.
- Engineered AWS Glue workflows to automate ingestion of S3-hosted transactional and budget data into Snowflake, cutting ETL runtime by 20% and providing near-real-time KPI availability.
- Built dynamic Excel dashboards using Pivot Tables and VLOOKUP, fed by Python-generated CSVs, to enable finance teams to access updated performance metrics within hours, accelerating decision cycles by 15%.
- Implemented a Pandas-based validation framework that automatically flagged missing values and inconsistencies pre-load, boosting report accuracy by 15% and eliminating manual reconciliation tasks.
- Integrated Scikit-learn models into the data pipeline, deploying regression for revenue forecasting and k-means clustering for cost-center segmentation, to guide quarterly budgeting and risk mitigation.
- Applied advanced SQL window functions and CTAS statements in Amazon Redshift to detect cost anomalies and extract multi-year trend insights, driving targeted efficiency initiatives.
- Designed live Tableau dashboards connected to Snowflake schemas, delivering interactive profitability and cost-trend visualizations that increased stakeholder adoption by 25%.
- Tuned Redshift performance by optimizing distribution and sort keys based on project query patterns, reducing average ad-hoc report runtimes by 25% and supporting faster analytics.

EDUCATION

Master of Science in Business Analytics and Project Management - University of Connecticut, Stamford, Connecticut, USA

Bachelor of Technology in Computer Science and Engineering - PVP Siddhartha Institute of Technology, Andhra Pradesh, India